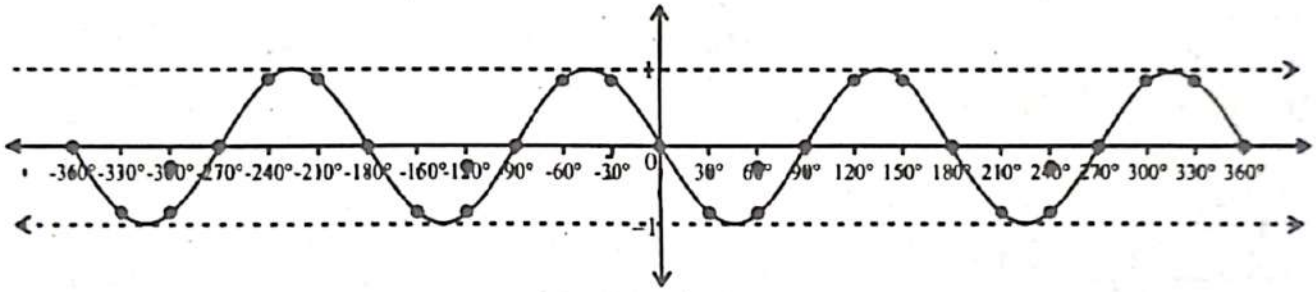
Exercise 11.2 SOLUTION

Q.1 Draw the graph of each of the following function for the intervals mentioned against each:

(i) $y = -\sin 2x, x \in [-2\pi, 2\pi]$

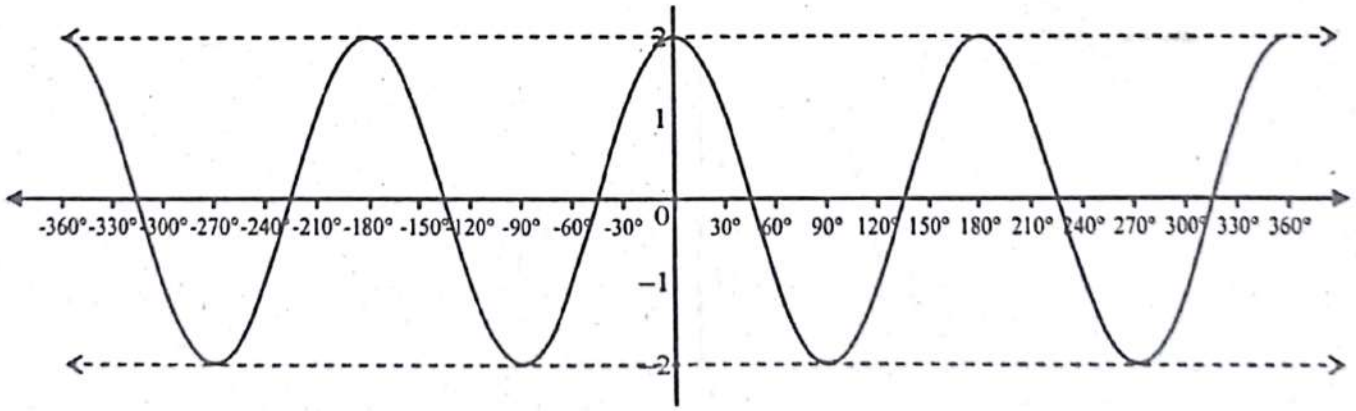
x	−360°	−300°	−240°	−180°	−120°	−60°	0°	60°	120°	180°	240°	300°	360°
y	0	0.866	−0.866	0	0.866	−0.866	0	0.866	−0.866	0	0.866	−0.866	0



$y = -\sin 2x$

(ii) $y = 2\cos 2x, x \in [-2\pi, 2\pi]$

x	−360°	−300°	−240°	−180°	−120°	−60°	0°	60°	120°	180°	240°	300°	360°
y	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1	2

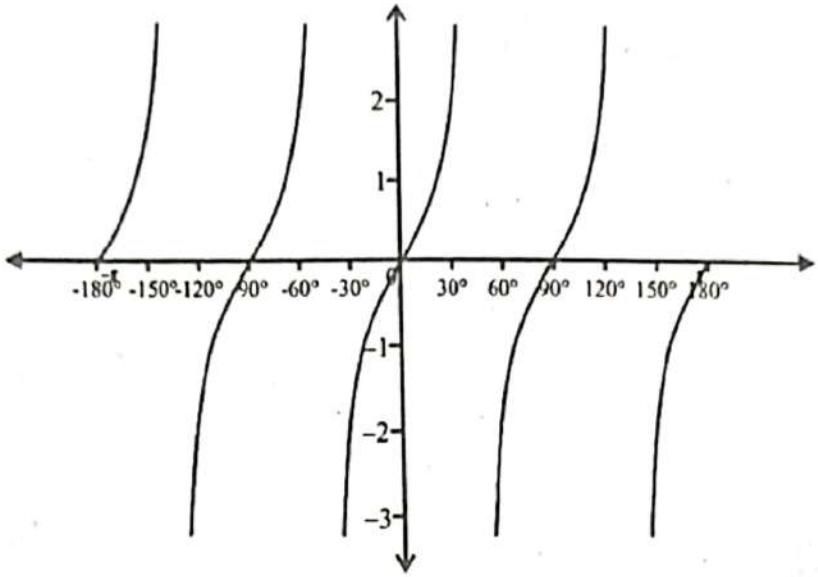


$y = 2\cos 2x$

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(iii) $y = \tan 2x, x \in [-\pi, \pi]$

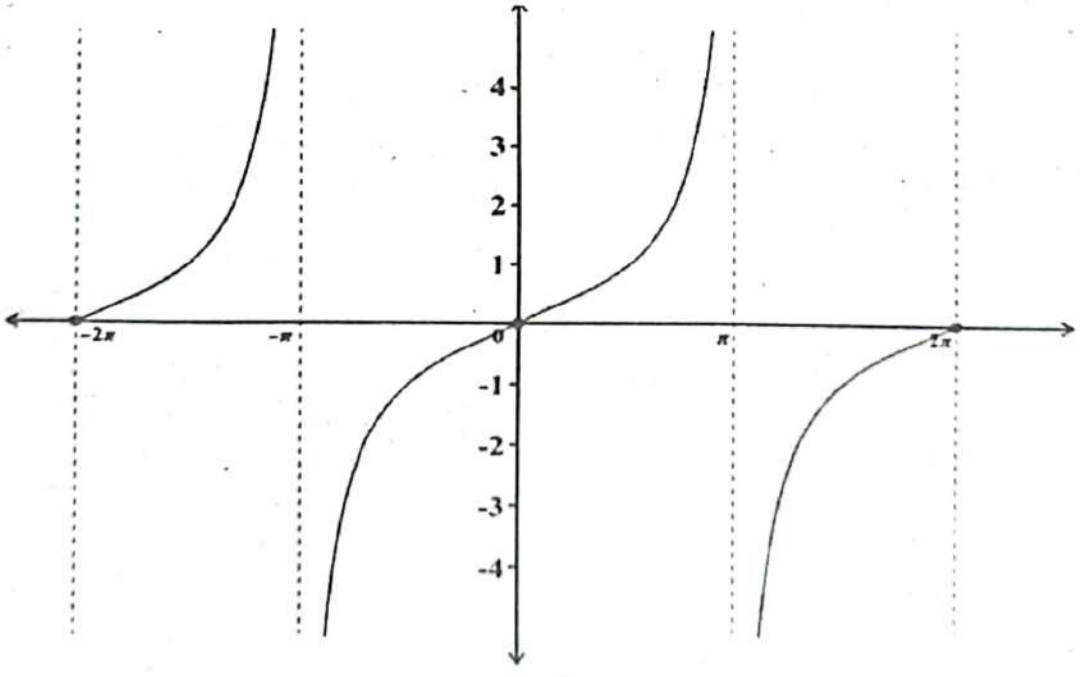
x	−180°	−120°	−60°	0°	60°	120°	180°
y	0	−1.732	1.732	0	−1.732	1.732	0



$y = \tan 2x$

(iv) $y = \tan \frac{x}{2}$, $x \in [-2\pi, 2\pi]$

x	−360°	−240°	−180°	−120°	−60°	0°	60°	120°	180°	240°	360°
y	0	1.732	∞	−1.732	−0.577	0	0.577	1.732	∞	−1.732	0

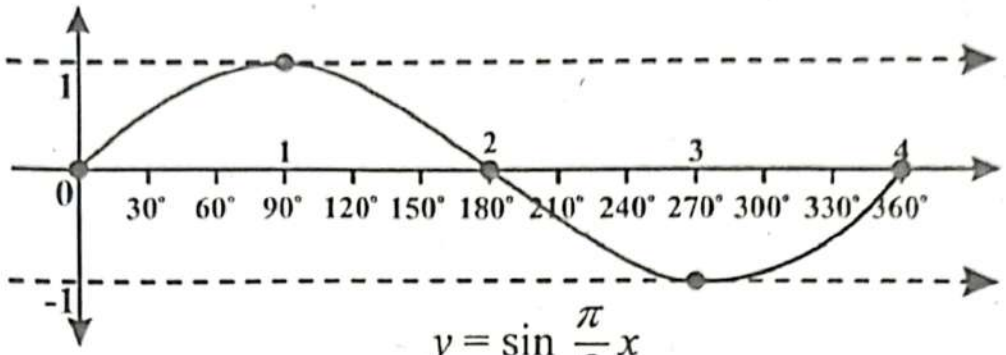


$y = \tan \frac{x}{2}$

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(v) $y = \sin \frac{\pi}{2} x \ , \ x \in [0, 2\pi]$

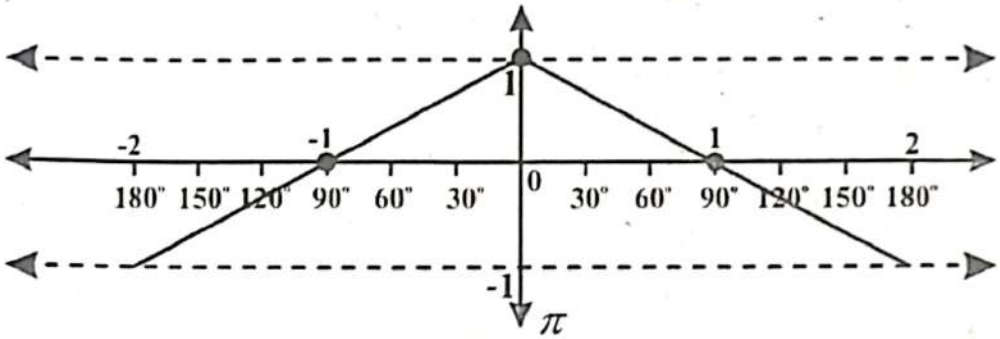
x	0	1	2	3	4
y	0	1	0	-1	0



$y = \sin \frac{\pi}{2} x$

(vi) $y = \cos \frac{\pi}{2} x \ , \ x \in [-\pi, \pi]$

x	-2	-1	0	1	2
y	-1	0	1	0	-1



$y = \cos \frac{\pi}{2} x$

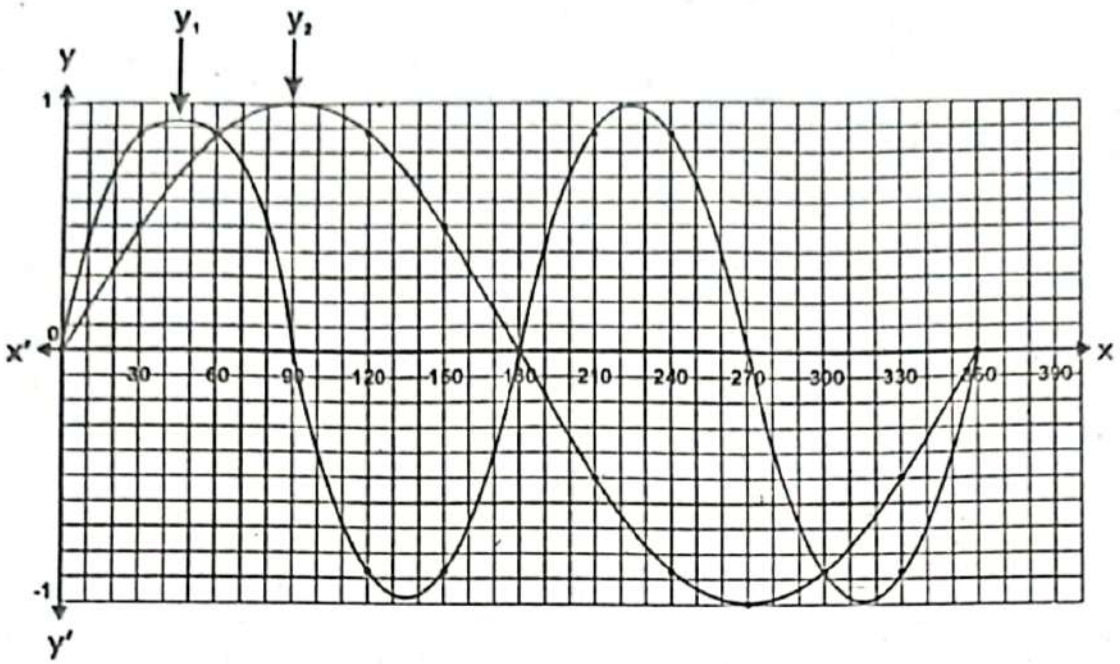
Q.2 On the same axes and to the same scale draw the graphs of the following functions for their complete period

i. $y = \sin x$ **and** $y = \sin 2x$

Sol: Take $y_1 = \sin x$ and $y_2 = \sin 2x$

x	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{6\pi}{6}$	$\frac{7\pi}{6}$	$\frac{8\pi}{6}$	$\frac{9\pi}{6}$	$\frac{10\pi}{6}$	$\frac{11\pi}{6}$	$\frac{2}{\pi}$
y_1	0	0.5	0.87	1	0.87	0.5	0	−0.5	−0.87	−1	−0.87	−0.5	0
y_2	0	0.87	0.87	0	−0.87	−0.87	0	0.87	0.87	0	−0.87	−0.87	0

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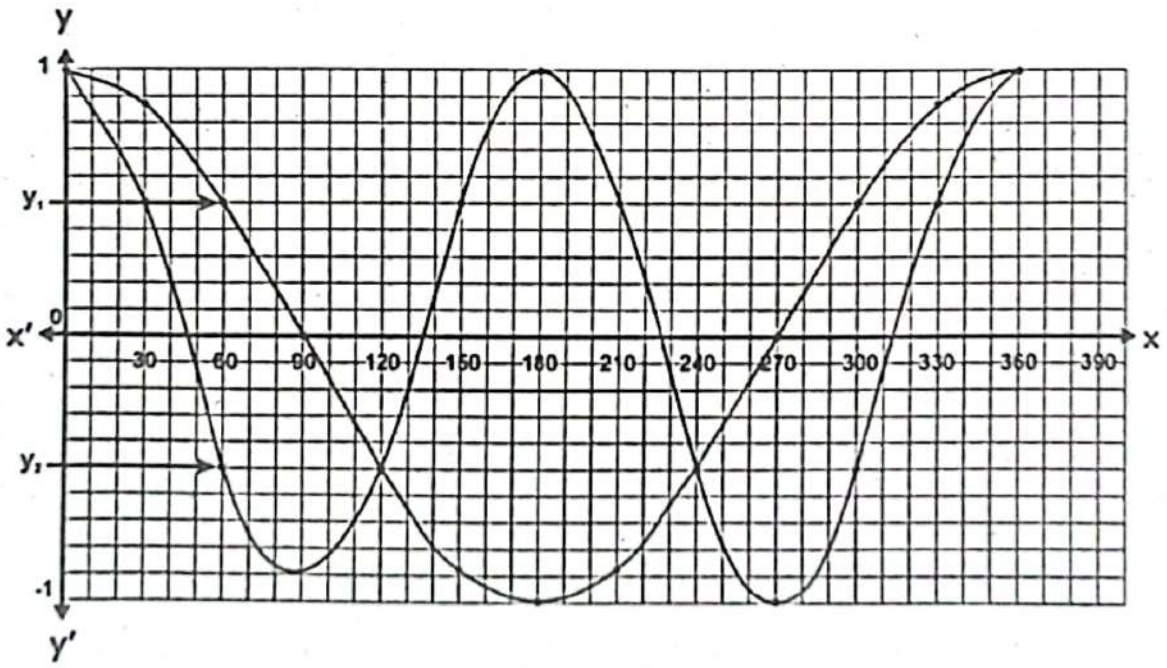
$y = \sin x$ *and* $y = \sin 2x$

ii. **y= cos x** **and** **y= cos2x**

Sol:

Take y_1=cosx and y_2=cos2x

X	0	60°	120°	180°	240°	300°	360°
Y_1	1	0.5	–0.5	–1	–0.5	0.5	1
Y_2	1	–0.5	–0.5	1	–0.5	–0.5	1



$y = \cos x$ *and* $y = \cos 2x$

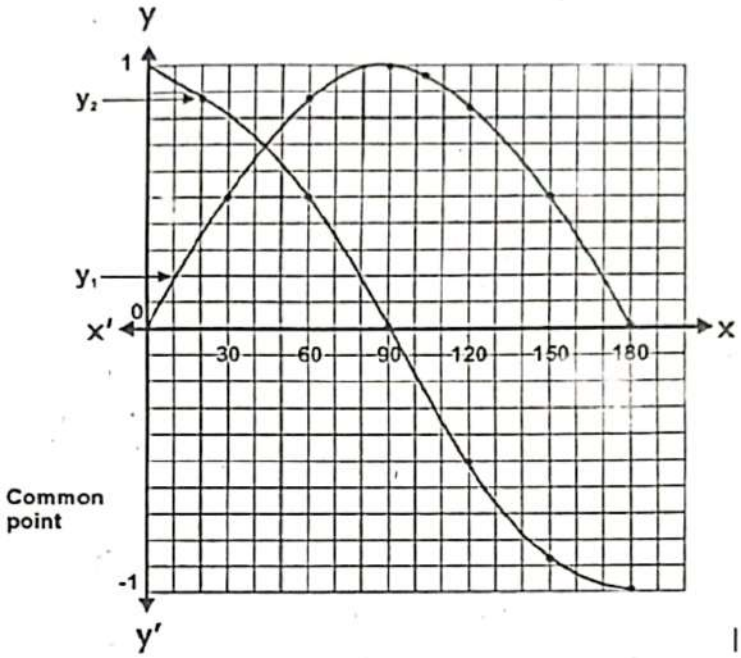
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3. Solve graphically

i. **Sin x = Cos x , $x \in [0, \pi]$**

Sol: Take $y_1 = \sin x$ $y_2 = \cos x$

X	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	π
Y_1	0	0.5	0.87	1	0.87	0.5	0
Y_2	1	0.866	0.5	0	−0.5	−0.866	−1



Point of intersection is at π/4 = 45°. Hence Solution set = { π/4 }

(ii). Sinx = x ; $x \in [0, \pi]$

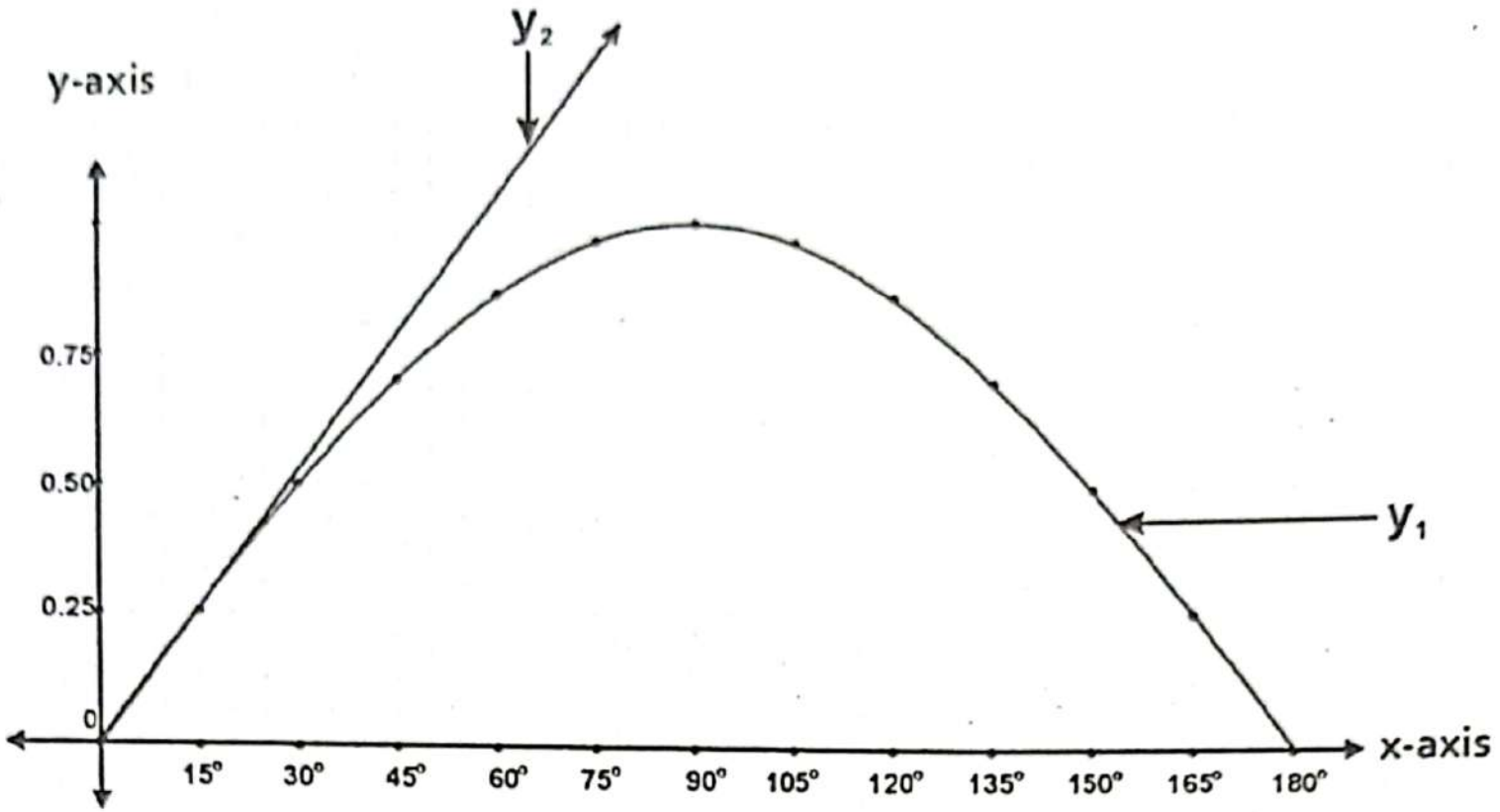
Let $y_1 = \sin x$ $y_2 = x$

Sol: $\{0\}$

x	0°	15°	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
$y_1 = \sin x$	0	0.25	0.50	0.70	0.87	0.96	1	0.96	0.87	0.70	0.50	0.25	0
$y_2 = x$	0	0.26	0.52	0.78	1.05	1.30	1.57	1.83	2.09	2.35	2.62	2.87	3.14

$$1^{c} = \frac{\pi}{180} = 0.01745$$

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y-axis
y_2
y_1
0.75
0.50
0.25
0
x-axis
15°
30°
45°
60°
75°
90°
105°
120°
135°
150°
165°
180°

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